Infosys_FarmVerse – Precision Agriculture Management Platform

Abstract

Agriculture plays a vital role in economic development, but farmers face challenges such as unpredictable weather, inefficient irrigation, soil degradation, and crop diseases. Traditional farming methods rely on manual monitoring, which often leads to excessive water usage and reduced crop productivity. IoTSys.FarmVerse is an IoT-based Precision Agriculture Management Platform that enables real-time monitoring of soil moisture, temperature, humidity, and environmental conditions. The platform helps farmers make informed decisions, optimize resource utilization, reduce operational costs, and improve crop yield through smart farming technologies.

Problem Statement

Agriculture is the backbone of many economies, yet farmers continue to face challenges such as unpredictable weather, inefficient irrigation, soil degradation, crop diseases, and the lack of real-time farm monitoring. Traditional farming methods often depend on manual observation, which can lead to excessive use of water, fertilizers, and pesticides, resulting in reduced crop yield and increased production costs. To address these issues, there is a need for a smart and technology-driven solution. IoTSys.FarmVerse – Precision Agriculture Management Platform aims to leverage IoT sensors, real-time data monitoring, and data analytics to help farmers make informed decisions. The platform enables continuous monitoring of soil moisture, temperature, humidity, and other environmental factors, allowing efficient resource utilization, timely crop management, and improved agricultural productivity. This solution supports sustainable farming practices while reducing operational costs and enhancing overall farm efficiency.

Proposed Solution

The proposed solution is IoTSys.FarmVerse – Precision Agriculture Management Platform, an IoT-based smart farming system designed to improve agricultural productivity and resource management. The platform collects real-time data from sensors measuring soil moisture, temperature, humidity, and other environmental conditions. This data is analyzed to provide farmers with timely recommendations for irrigation, crop health monitoring, and fertilizer usage. The system also offers a user-friendly dashboard where farmers can monitor farm conditions remotely and receive alerts about critical issues. By integrating IoT technology with data analytics, the platform enables informed decision-making, reduces water and resource wastage, lowers operational costs, and promotes sustainable farming practices, ultimately improving crop yield and overall farm efficiency.

Objectives of the Project

- To develop an IoT-based Precision Agriculture Management Platform for smart farm monitoring.
- To monitor soil moisture, temperature, humidity, and other environmental conditions in real time using IoT sensors.
- To provide timely irrigation recommendations to reduce water wastage and improve water management.
- To detect potential crop and environmental issues early through continuous monitoring and alerts.
- To help farmers make data-driven decisions for better crop growth and higher productivity.
- To optimize the use of fertilizers and other agricultural resources, reducing operational costs.
- To provide a user-friendly web platform for remote monitoring and farm management.
- To promote sustainable and efficient farming practices through the use of modern technologies.

System Architecture

IoTSys.FarmVerse
Precision Agriculture Management Platform

🧑‍🌾 Farmer / User
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React Web Application
(HTML, CSS, Bootstrap)
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HTTP Requests
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▼
Spring Boot REST Controller
(Spring MVC)
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Business Logic Layer
(Spring Boot Services)
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▼
Spring Data JPA Repository
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MySQL Database
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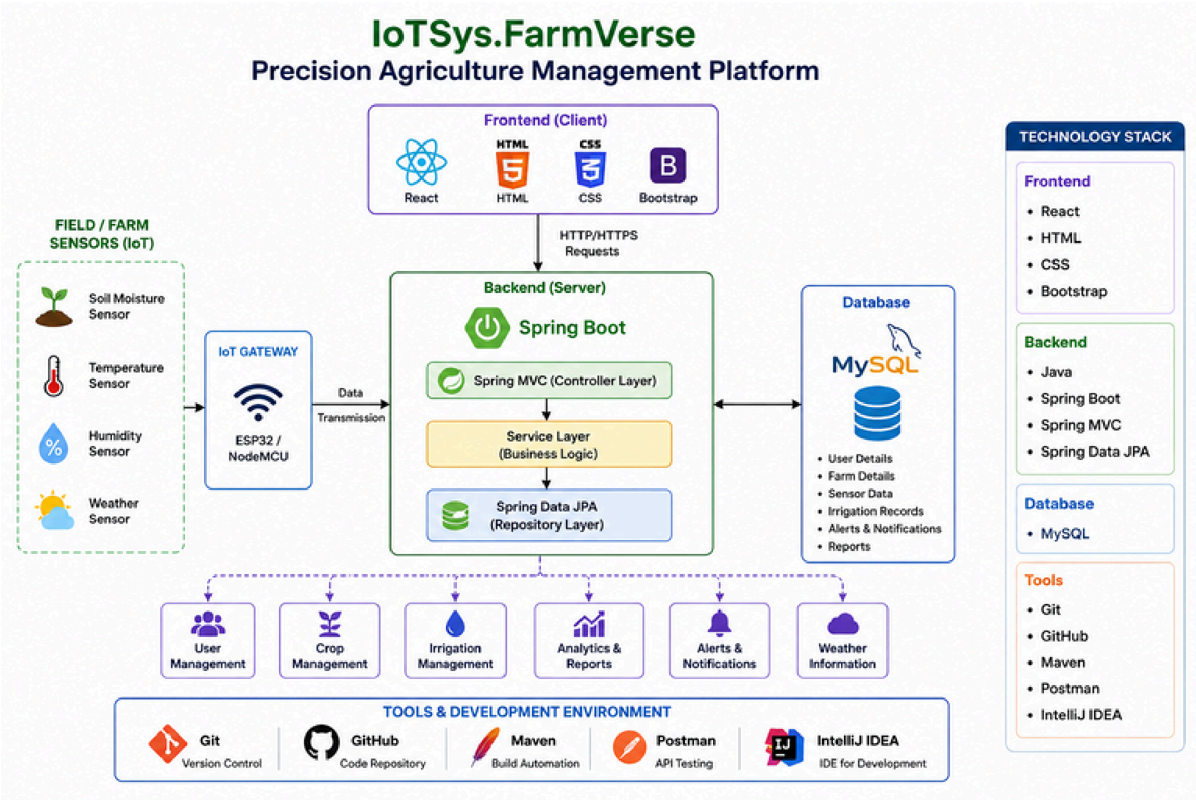
Sensor Data / Farm Information
(Soil Moisture, Temperature, Humidity)



Existing System

- Manual monitoring of farms.
- Excessive water usage.
- Delayed identification of crop diseases.
- Lack of real-time environmental monitoring.
- Low productivity.
- High labour dependency.

Proposed System

- IoT sensor-based monitoring.
- Automated environmental data collection.
- Remote farm monitoring.
- Smart irrigation recommendations.
- Real-time alerts.
- Better crop management.
- Higher productivity.

Work Flow

- User logs into the FarmVerse web application.
- The farmer enters or updates farm details.
- IoT sensors collect real-time data such as soil moisture, temperature, humidity, and weather conditions.
- The collected sensor data is transmitted to the backend server through the IoT gateway (ESP32/NodeMCU).
- The Spring Boot backend processes and validates the received data.
- The processed information is stored in the MySQL database using Spring Data JPA.
- The system analyzes the data to determine crop conditions and irrigation requirements.
- The React frontend retrieves the processed data through REST APIs.
- The dashboard displays real-time farm status, reports, and analytics to the user.
- The system generates alerts and recommendations for irrigation, crop health, and weather conditions.
- The farmer uses these recommendations to make informed decisions, improving productivity and reducing resource wastage.

Modules Developed

- User Authentication – Login and registration.
- Farm Management – Manage farm and crop details.
- Sensor Monitoring – Collect and display IoT sensor data.
- Irrigation Management – Smart irrigation recommendations.
- Weather Monitoring – Display weather information.
- Dashboard – View real-time farm status and analytics.
- Alerts & Notifications – Notify users about important updates.
- Reports – Generate farm performance reports.

Technology Stack

Frontend

- React
- HTML5
- CSS3
- Bootstrap

Backend

- Java
- Spring Boot
- Spring MVC
- Spring Data JPA

Tools

- IntelliJ IDEA
- Visual Studio Code (VS Code)
- Git
- GitHub
- Maven
- Postman

Database

- MySQL

Future Scope

- AI-based crop disease prediction.
- Drone-based farm monitoring.
- Mobile application.
- Weather forecasting integration.
- Automatic irrigation control.
- Fertilizer recommendation using AI.